Dynamic Phasor Model

I. IEEE 9 BUS SYSTEM

A. State-Space Model

The transmission line in IEEE 9 bus system can be modeled in the time domain as follows:

$$\begin{cases} \frac{di_{12}}{dt} = \frac{1}{L} \cdot (v_1 - v_2 - i_{12} \cdot R), \\ \frac{dv_1}{dt} = \frac{1}{C_1} \cdot (i_1 - i_{12}), \\ \frac{dv_2}{dt} = \frac{1}{C_2} \cdot (i_{12} - i_2), \end{cases} \tag{1}$$

With regard to the generator-connected bus, the time domain based model can be established as follows:

$$\begin{cases} \frac{di_{14}}{dt} = \frac{1}{L_1 + L_{14}} \cdot (v_{s1} - v_4 - i_{14} \cdot R_{14}), \\ \frac{dv_4}{dt} = \frac{1}{C_4} \cdot (i_{14} - i_{49} - i_{45}), \end{cases} \tag{2}$$

Therefore, the state-space equations of the whole system can be established as follows:

$$\frac{d\boldsymbol{x}(t)}{dt} = A\boldsymbol{x}(t) + B\boldsymbol{u}(t), \tag{3}$$

With regard to its dynamic phasor model, the product feature and the differential feature are given as follows:

$$\begin{cases} (\widehat{\boldsymbol{xy}})_{\boldsymbol{k}} = \sum_i \hat{\boldsymbol{x}}_{\boldsymbol{k}} - \boldsymbol{i}\hat{\boldsymbol{y}}_{\boldsymbol{i}}, \\ \frac{d}{dt}\hat{\boldsymbol{x}}_{\boldsymbol{k}} = (\widehat{\frac{\boldsymbol{dx}}{\boldsymbol{dt}}})_{\boldsymbol{k}} - jk\omega\hat{\boldsymbol{x}}_{\boldsymbol{k}}, \end{cases} \tag{4}$$

Therefore, the corresponding dynamic phasor model of eq. (3) can be derived as follows:

$$\frac{d\hat{\boldsymbol{x}}_{\boldsymbol{k}}(t)}{dt} = A_k\hat{\boldsymbol{x}}_{\boldsymbol{k}}(t) + B\hat{\boldsymbol{u}}_{\boldsymbol{k}}(t), \tag{5}$$

where $A_k = A - jk\omega I$.

For simplicity, time variable t is omitted and eq. (5) can be rewritten in the real and imaginary part separately as follows:

$$\frac{d}{dt}\begin{bmatrix} \hat{\boldsymbol{x}}_{\boldsymbol{R,k}} \\ \hat{\boldsymbol{x}}_{\boldsymbol{I,k}} \end{bmatrix} = \begin{bmatrix} A & k\omega I \\ -k\omega I & A \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{x}}_{\boldsymbol{R,k}} \\ \hat{\boldsymbol{x}}_{\boldsymbol{I,k}} \end{bmatrix} + \begin{bmatrix} B & O \\ O & B \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{u}}_{\boldsymbol{R,k}} \\ \hat{\boldsymbol{u}}_{\boldsymbol{I,k}}, \end{bmatrix} \tag{6}$$

where $\hat{\boldsymbol{x}}_{\boldsymbol{R,k}} = Re(\hat{\boldsymbol{x}}_{\boldsymbol{k}})$, $\hat{\boldsymbol{x}}_{\boldsymbol{I,k}} = Im(\hat{\boldsymbol{x}}_{\boldsymbol{k}})$, $\hat{\boldsymbol{u}}_{\boldsymbol{R,k}} = Re(\hat{\boldsymbol{u}}_{\boldsymbol{k}})$, $\hat{\boldsymbol{u}}_{\boldsymbol{I,k}} = Im(\hat{\boldsymbol{u}}_{\boldsymbol{k}})$,

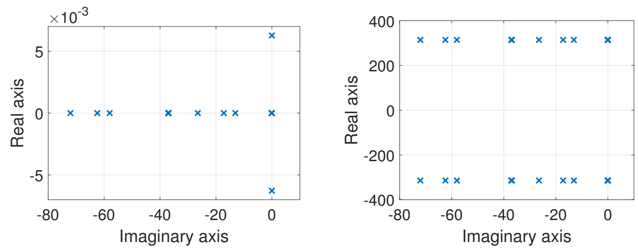
There is another way which can be used to convert the time-domain based model into the dynamic phasor based model.

Take eq. (1) as instance, the time-domain based model in eq. (1) can be firstly converted into the complex domain as follows:

$$\begin{cases} s \cdot \hat{I}_{12}(s) = \frac{1}{L} \cdot [\hat{V}_1(s) - \hat{V}_2(s) - \hat{I}_{12}(s) \cdot R], \\ s \cdot \hat{V}_1(s) = \frac{1}{C_1} \cdot [\hat{I}_1(s) - \hat{I}_{12}(s)], \\ s \cdot \hat{V}_2(s) = \frac{1}{C_2} \cdot [\hat{I}_{12}(s) - \hat{I}_2(s)], \end{cases} \tag{7}$$

With regard to the k^{th} dynamic phasor, we can define

$$\begin{cases} \hat{I}_{12}(s) = \hat{I}_{12,Re,k} + j \cdot \hat{I}_{12,Im,k}, \\ \hat{V}_1(s) = \hat{V}_{1,Re,k} + j \cdot \hat{V}_{1,Im,k}, \\ \hat{V}_2(s) = \hat{V}_{2,Re,k} + j \cdot \hat{V}_{2,Im,k}, \end{cases} \tag{8}$$



(a) (b)

Figure 1. Eigen values.

We can also define $s = p + j \cdot \omega$, where $p = \frac{d}{dt}$. Then we can derive the dynamic phase model of eq. (1), which is same with eq. (6).

B. Stability Analysis

The model stability can be investigated based on the eigenvalues of matrix A. With regard to the time-domain model in eq. (3), the eigenvalues of matrix A can be illustrated in the Fig. 1.

As shown, all eigenvalues have negative real parts, hence both the time-domain model and the dynamic phasor model are stable.

II. LCL CONVERTER

A. State-Space Model

The circuit equations in the time-domain can be described as follows:

$$\begin{cases} \frac{di_{gA}}{dt} = \frac{v_{pccA} - v_{cfA}}{L_g} - \frac{R_{gA}}{L_{gA}} i_{gA}, \\ \frac{di_{gB}}{dt} = \frac{v_{pccB} - v_{cfB}}{L_g} - \frac{R_{gB}}{L_{gB}} i_{gB}, \\ \frac{di_{gC}}{dt} = \frac{v_{pccC} - v_{cfC}}{L_g} - \frac{R_{gC}}{L_{gC}} i_{gC}, \\ \frac{di_{fA}}{dt} = \frac{v_{cfA} - v_{invA}}{L_{fA}} - \frac{R_{fA}}{L_{fA}} i_{fA}, \\ \frac{di_{fB}}{dt} = \frac{v_{cfB} - v_{invB}}{L_{fB}} - \frac{R_{fB}}{L_{fB}} i_{fB}, \\ \frac{di_{fC}}{dt} = \frac{v_{cfC} - v_{invC}}{L_{fC}} - \frac{R_{fC}}{L_{fC}} i_{fC}, \\ \frac{dv_{capA}}{dt} = \frac{1}{C_{fA}} \cdot (i_{gA} - i_{fA}), \\ \frac{dv_{capB}}{dt} = \frac{1}{C_{fB}} \cdot (i_{gB} - i_{fB}), \\ \frac{dv_{capC}}{dt} = \frac{1}{C_{fC}} \cdot (i_{gC} - i_{fC}), \\ \frac{dv_{dc}}{dt} = \frac{1}{C_{dc}} i_{dc} - \frac{1}{R_{dc}C_{dc}} v_{dc}, \end{cases} \tag{9}$$

where

$$\begin{cases} v_{cfA} = v_{capA} + (i_{gA} - i_{fA}) \cdot R_{cfA}, \\ v_{cfB} = v_{capB} + (i_{gB} - i_{fA}) \cdot R_{cfB}, \\ v_{cfC} = v_{capC} + (i_{gC} - i_{fC}) \cdot R_{cfC}, \\ v_{invA} = v_{dc} \cdot sw_A, \\ v_{invB} = v_{dc} \cdot sw_B, \\ v_{invC} = v_{dc} \cdot sw_C, \\ i_{dc} = i_{fA} \cdot sw_A + i_{fB} \cdot sw_B + i_{fC} \cdot sw_C, \end{cases} \tag{10}$$

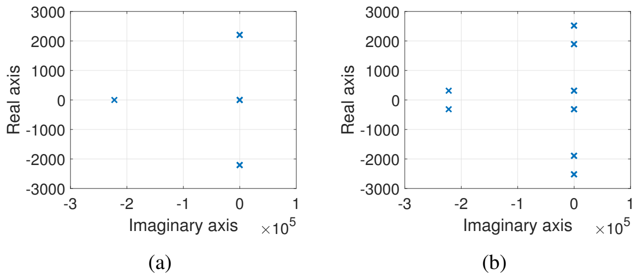


Figure 2. Eigen values.

the small-signal model of terms $v_{invA} = v_{dc}Sw_A$ can be expressed as follows:

$$\begin{aligned} V_{invA0} + \Delta v_{invA} &= (Sw_{A0} + \Delta sw_A)(V_{dc0} + \Delta v_{dc}) \\ &= Sw_{A0} \cdot V_{dc0} + Sw_{A0} \cdot \Delta v_{dc} \\ &+ V_{dc0} \cdot \Delta sw_{A0} + \Delta sw_{A0} \cdot \Delta v_{dc}, \end{aligned} \tag{11}$$

hence we obtain

$$\Delta v_{invA} = Sw_{A0} \cdot \Delta v_{dc} + V_{dc0} \cdot \Delta sw_{A0}. \tag{12}$$

Similarly, we can obtain the following equations

$$\begin{cases} \Delta v_{invB} &= Sw_{B0} \cdot \Delta v_{dc} + V_{dc0} \cdot \Delta sw_B, \\ \Delta v_{invC} &= Sw_{C0} \cdot \Delta v_{dc} + V_{dc0} \cdot \Delta sw_C, \\ \Delta i_{dc} &= Sw_{A0} \cdot \Delta i_{fA} + i_{fA0} \cdot \Delta sw_A \\ &+ Sw_{B0} \cdot \Delta i_{fB} + i_{fB0} \cdot \Delta sw_B \\ &+ Sw_{C0} \cdot \Delta i_{fC} + i_{fC0} \cdot \Delta sw_C, \end{cases} \tag{13}$$

Therefore, we can obtain the small-signal model as follows:

$$\begin{cases} \frac{d\Delta i_{gA}}{dt} &= (-\frac{R_{gA}}{L_{gA}} - \frac{R_{cfA}}{L_{gA}})\Delta i_{gA} - \frac{1}{L_{gA}}\Delta v_{capA} \\ &+ \frac{R_{cfA}}{L_{gA}}\Delta i_{fA} + \frac{1}{L_{gA}}\Delta v_{pccA}, \\ \frac{d\Delta i_{gB}}{dt} &= (-\frac{R_{gB}}{L_{gB}} - \frac{R_{cfB}}{L_{gB}})\Delta i_{gB} - \frac{1}{L_{gB}}\Delta v_{capB} \\ &+ \frac{R_{cfB}}{L_{gB}}\Delta i_{fB} + \frac{1}{L_{gB}}\Delta v_{pccB}, \\ \frac{d\Delta i_{gC}}{dt} &= (-\frac{R_{gC}}{L_{gC}} - \frac{R_{cfC}}{L_{gC}})\Delta i_{gC} - \frac{1}{L_{gC}}\Delta v_{capC} \\ &+ \frac{R_{cfC}}{L_{gC}}\Delta i_{fC} + \frac{1}{L_{gC}}\Delta v_{pccC}, \\ \frac{d\Delta i_{fA}}{dt} &= \frac{R_{cfA}}{L_{fA}}\Delta i_{gA} + (-\frac{R_{cfA}+R_{fA}}{L_{fA}})\Delta i_{fA} \\ &+ \frac{1}{L_{fA}}\Delta v_{capA} + \frac{-Sw_{A0}}{L_{fA}}\Delta v_{dc} + \frac{-V_{dc0}}{L_{fA}}\Delta sw_A \\ \frac{d\Delta i_{fB}}{dt} &= \frac{R_{cfB}}{L_{fB}}\Delta i_{gB} + (-\frac{R_{cfB}+R_{fB}}{L_{fB}})\Delta i_{fB} \\ &+ \frac{1}{L_{fB}}\Delta v_{capB} + \frac{-Sw_{B0}}{L_{fB}}\Delta v_{dc} + \frac{-V_{dc0}}{L_{fB}}\Delta sw_B \\ \frac{d\Delta i_{fC}}{dt} &= \frac{R_{cfC}}{L_{fC}}\Delta i_{gC} + (-\frac{R_{cfC}+R_{fC}}{L_{fC}})\Delta i_{fC} \\ &+ \frac{1}{L_{fC}}\Delta v_{capC} + \frac{-Sw_{A0}}{L_{fC}}\Delta v_{dc} + \frac{-V_{dc0}}{L_{fC}}\Delta sw_C \\ \frac{d\Delta v_{capA}}{dt} &= \frac{1}{C_{fA}}\Delta i_{gA} - \frac{1}{C_{fA}}\Delta i_{fA} \\ \frac{d\Delta v_{capB}}{dt} &= \frac{1}{C_{fB}}\Delta i_{gB} - \frac{1}{C_{fB}}\Delta i_{fB} \\ \frac{d\Delta v_{capC}}{dt} &= \frac{1}{C_{fC}}\Delta i_{gC} - \frac{1}{C_{fC}}\Delta i_{fC} \\ \frac{d\Delta v_{dc}}{dt} &= \frac{Sw_{A0}}{C_{dc}}\Delta i_{fA} + \frac{Sw_{B0}}{C_{dc}}\Delta i_{fB} + \frac{Sw_{C0}}{C_{dc}}\Delta i_{fC} \\ &+ \frac{i_{fA0}}{C_{dc}}\Delta sw_A + \frac{i_{fB0}}{C_{dc}}\Delta sw_B + \frac{i_{fC0}}{C_{dc}}\Delta sw_C \\ &- \frac{1}{R_{dc}C_{dc}}\Delta v_{dc} \end{cases} \tag{14}$$

B. Stability Analysis

III. CONTROLLER

The transfer function of the PR controller can be expressed as follows:

$$PR(s) = k_p + \frac{2k_r\omega_c s}{s^2 + 2\omega_c s + \omega_0^2}. \tag{15}$$

We define $k_1 = k_p, k_2 = 2k_r\omega_c, k_3 = 2\omega_c$, then time-domain based state space equations of PR controller can be expressed as follows:

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & -k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \tag{16}$$

$$y = \begin{bmatrix} 0 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + k_1 u \tag{17}$$

Since we convert the abc coordinate to the $\alpha - \beta$ coordinate in the PR controller, there will be four state variables, including $x_{1,\alpha}, x_{2,\alpha}, x_{1,\beta}, x_{2,\beta}$.

With regard to the input variables, we require $i_{fX}(X = A, B, C), i_{gX}(X = A, B, C), i_{g\alpha,ref}, i_{g\beta,ref}$, hence the state equations can be written as follows:

$$\begin{cases} \dot{x_1} &= x_2, \\ \dot{x_2} &= -\omega_0^2 x_1 - k_3 x_2 + (\frac{2}{3}i_{gA} - \frac{1}{3}i_{gB} - \frac{1}{3}i_{gC}) + i_{g\alpha,ref}, \\ \dot{x_3} &= x_4, \\ \dot{x_4} &= -\omega_0^2 x_3 - k_3 x_4 + (\frac{1}{\sqrt{3}}i_{gB} - \frac{1}{\sqrt{3}}i_{gC}) + i_{g\beta,ref}, \end{cases} \tag{18}$$

Its matrix form can be written as follows:

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{x_3} \\ \dot{x_4} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\omega_0^2 & -k_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega_0^2 & -k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \tag{19}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{g\alpha,ref} \\ i_{g\beta,ref} \\ i_{gA} \\ i_{gB} \\ i_{gC} \\ i_{fA} \\ i_{fB} \\ i_{fC} \end{bmatrix} \tag{20}$$

With regard to the output equations, they can be written as follows:

$$\begin{cases} u_\alpha &= \frac{2}{3}i_{gA} - \frac{1}{3}i_{gB} - \frac{1}{3}i_{gC} + i_{g\alpha,ref}, \\ u_\beta &= \frac{1}{\sqrt{3}}i_{gB} - \frac{1}{\sqrt{3}}i_{gC} + i_{g\beta,ref}, \\ y_\alpha &= k_2 x_2 + k_1 u_\alpha + \\ & k_d[-\frac{2}{3}(i_{gA} - i_{fA}) + \frac{1}{3}(i_{gB} - i_{fB}) + \frac{1}{3}(i_{gC} - i_{fC})], \\ y_\beta &= k_2 x_4 + k_1 u_\beta + \\ & k_d[-\frac{1}{\sqrt{3}}(i_{gB} - i_{fB}) + \frac{1}{\sqrt{3}}(i_{gC} - i_{fC})] \end{cases} \tag{21}$$

In addition, the output variables can be obtained according to the Clark inverse transform as follows:

$$\begin{bmatrix} S_A \\ S_B \\ S_C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} y_\alpha \\ y_\beta \end{bmatrix} \tag{22}$$

Therefore, the matrix form of the output equations can be expressed as follows:

APPENDIX

$$
\begin{bmatrix} S_A \\ S_B \\ S_C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} k_1 & 0 & (k_1-k_d)(\frac{2}{3}) & (k_1-k_d)(-\frac{1}{3}) & (k_1-k_d)(-\frac{1}{3}) & k_d(\frac{2}{3}) & k_d(-\frac{1}{3}) & k_d(-\frac{1}{3}) \\ k_1(-\frac{1}{2}) & k_1(\frac{\sqrt{3}}{2}) & (k_1-k_d)(-\frac{1}{3}) & (k_1-k_d)(\frac{2}{3}) & (k_1-k_d)(-\frac{1}{3}) & k_d(-\frac{1}{3}) & k_d(\frac{2}{3}) & k_d(-\frac{1}{3}) \\ k_1(-\frac{1}{2}) & k_1(-\frac{\sqrt{3}}{2}) & (k_1-k_d)(-\frac{1}{3}) & (k_1-k_d)(-\frac{1}{3}) & (k_1-k_d)(\frac{2}{3}) & k_d(-\frac{1}{3}) & k_d(-\frac{1}{3}) & k_d(-\frac{2}{3}) \end{bmatrix} \begin{bmatrix} i_{g\alpha,ref} \\ i_{g\beta,ref} \\ i_{gA} \\ i_{gB} \\ i_{gC} \\ i_{fA} \\ i_{fB} \\ i_{fC} \end{bmatrix} \tag{23}
$$

$$
A = \begin{bmatrix}
-\frac{R_{14}}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{R_{49}}{L_{49}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{49}} & 0 & 0 & 0 & 0 & -\frac{1}{L_{49}} \\
0 & 0 & -\frac{R_{45}}{L_{45}} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{45}} & -\frac{1}{L_{45}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{R_{89}}{L_{89}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{89}} & -\frac{1}{L_{89}} \\
0 & 0 & 0 & 0 & -\frac{R_{65}}{L_{65}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{65}} & \frac{1}{L_{65}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{R_{36}}{L_{36}+L_3} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{36}+L_3} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{67}}{L_{67}} & 0 & 0 & 0 & 0 & \frac{1}{L_{67}} & -\frac{1}{L_{67}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{87}}{L_{87}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{87}} & \frac{1}{L_{87}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{28}}{L_{28}+L_2} & 0 & 0 & 0 & 0 & -\frac{1}{L_{28}+L_2} & 0 \\
\frac{1}{C_4} & -\frac{1}{C_4} & -\frac{1}{C_4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{C_5} & 0 & \frac{1}{C_5}0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{1}{C_6} & \frac{1}{C_6} & -\frac{1}{C_6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_7} & \frac{1}{C_7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{1}{C_8} & 0 & 0 & -\frac{1}{C_8} & \frac{1}{C_8} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{C_9} & 0 & \frac{1}{C_9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix},
$$

$$
B = \begin{bmatrix}
\frac{1}{L_{14}+L_1} & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & \frac{1}{L_{36}+L_3} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & \frac{1}{L_{28}+L_2} & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}, \boldsymbol{x}(t) = \begin{bmatrix} i_{14} \\ i_{49} \\ i_{45} \\ i_{89} \\ i_{65} \\ i_{36} \\ i_{67} \\ i_{87} \\ i_{28} \\ v_4 \\ v_5 \\ v_6 \\ v_7 \\ v_8 \\ v_9 \end{bmatrix}, \boldsymbol{u}(t) = \begin{bmatrix} v_{s1} \\ v_{s2} \\ v_{s3} \end{bmatrix}
$$

$$
\begin{bmatrix} A & k\omega I \\ -k\omega I & A \end{bmatrix} = \left[\begin{smallmatrix}
-\frac{R_{14}}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{R_{49}}{L_{49}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{49}} & 0 & 0 & 0 & 0 & -\frac{1}{L_{49}} & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{R_{45}}{L_{45}} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{45}} & -\frac{1}{L_{45}} & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{R_{89}}{L_{89}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{89}} & -\frac{1}{L_{89}} & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{R_{65}}{L_{65}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{65}} & \frac{1}{L_{65}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{R_{36}}{L_{36}+L_3} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{36}+L_3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{67}}{L_{67}} & 0 & 0 & 0 & 0 & \frac{1}{L_{67}} & -\frac{1}{L_{67}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{87}}{L_{87}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{87}} & \frac{1}{L_{87}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{28}}{L_{28}+L_2} & 0 & 0 & 0 & 0 & -\frac{1}{L_{28}+L_2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{C_4} & -\frac{1}{C_4} & -\frac{1}{C_4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{C_5} & 0 & \frac{1}{C_5}0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{1}{C_6} & \frac{1}{C_6} & -\frac{1}{C_6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_7} & \frac{1}{C_7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{1}{C_8} & 0 & 0 & -\frac{1}{C_8} & \frac{1}{C_8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 \\
0 & \frac{1}{C_9} & 0 & \frac{1}{C_9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega \\
-k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{14}}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 \\
0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{49}}{L_{49}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{49}} & 0 & 0 & 0 & 0 & -\frac{1}{L_{49}} \\
0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{45}}{L_{45}} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{45}} & -\frac{1}{L_{45}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{89}}{L_{89}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{89}} & -\frac{1}{L_{89}} \\
0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{65}}{L_{65}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{65}} & \frac{1}{L_{65}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{36}}{L_{36}+L_3} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{36}+L_3} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{67}}{L_{67}} & 0 & 0 & 0 & 0 & \frac{1}{L_{67}} & -\frac{1}{L_{67}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{87}}{L_{87}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_{87}} & \frac{1}{L_{87}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{R_{28}}{L_{28}+L_2} & 0 & 0 & 0 & 0 & -\frac{1}{L_{28}+L_2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & \frac{1}{C_4} & -\frac{1}{C_4} & -\frac{1}{C_4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{C_5} & 0 & \frac{1}{C_5}0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_6} & \frac{1}{C_6} & -\frac{1}{C_6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_7} & \frac{1}{C_7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_8} & 0 & 0 & -\frac{1}{C_8} & \frac{1}{C_8} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & \frac{1}{C_9} & 0 & \frac{1}{C_9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{smallmatrix}\right],
$$

$$
\begin{bmatrix} B & O \\ O & B \end{bmatrix} = \begin{bmatrix}
\frac{1}{L_{14}+L_1} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{L_{36}+L_3} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{L_{28}+L_2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{L_{14}+L_1} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{L_{36}+L_3} \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{L_{28}+L_2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}, \begin{bmatrix} \hat{\boldsymbol{x}}_{\boldsymbol{R,k}} \\ \hat{\boldsymbol{x}}_{\boldsymbol{I,k}} \end{bmatrix} = \begin{bmatrix}
\hat{i}_{14,R,k} \\ \hat{i}_{49,R,k} \\ \hat{i}_{45,R,k} \\ \hat{i}_{28,R,k} \\ \hat{i}_{87,R,k} \\ \hat{i}_{89,R,k} \\ \hat{i}_{36,R,k} \\ \hat{i}_{67,R,k} \\ \hat{i}_{65,R,k} \\ \hat{v}_{4,R,k} \\ \hat{v}_{5,R,k} \\ \hat{v}_{6,R,k} \\ \hat{v}_{7,R,k} \\ \hat{v}_{8,R,k} \\ \hat{v}_{9,R,k} \\ \hat{i}_{14,I,k} \\ \hat{i}_{49,I,k} \\ \hat{i}_{45,I,k} \\ \hat{i}_{28,I,k} \\ \hat{i}_{87,I,k} \\ \hat{i}_{89,I,k} \\ \hat{i}_{36,I,k} \\ \hat{i}_{67,I,k} \\ \hat{i}_{65,I,k} \\ \hat{v}_{4,I,k} \\ \hat{v}_{5,I,k} \\ \hat{v}_{6,I,k} \\ \hat{v}_{7,I,k} \\ \hat{v}_{8,I,k} \\ \hat{v}_{9,I,k}
\end{bmatrix}, \begin{bmatrix} \hat{\boldsymbol{u}}_{\boldsymbol{R,k}} \\ \hat{\boldsymbol{u}}_{\boldsymbol{I,k}} \end{bmatrix} = \begin{bmatrix}
\hat{v}_{s1,R,k} \\ \hat{v}_{s2,R,k} \\ \hat{v}_{s3,R,k} \\ \hat{v}_{s1,I,k} \\ \hat{v}_{s2,I,k} \\ \hat{v}_{s3,I,k}
\end{bmatrix}
$$

$$A = \begin{bmatrix}
-\frac{-R_{gA}-R_{cfA}}{L_{gA}} & 0 & 0 & \frac{-1}{L_{gA}} & 0 & 0 & \frac{R_{cfA}}{L_{gA}} & 0 & 0 & 0 \\
0 & -\frac{-R_{gB}-R_{cfB}}{L_{gB}} & 0 & 0 & \frac{-1}{L_{gB}} & 0 & 0 & \frac{R_{cfB}}{L_{gB}} & 0 & 0 \\
0 & 0 & -\frac{-R_{gC}-R_{cfC}}{L_{gC}} & 0 & 0 & \frac{-1}{L_{gC}} & 0 & 0 & \frac{R_{cfC}}{L_{gC}} & 0 \\
\frac{1}{C_{fA}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fA}} & 0 & 0 & 0 \\
0 & \frac{1}{C_{fB}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fB}} & 0 & 0 \\
0 & 0 & \frac{1}{C_{fC}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fC}} & 0 \\
\frac{R_{cfA}}{L_{fA}} & 0 & 0 & \frac{1}{L_{fA}} & 0 & 0 & -\frac{R_{cfA}-R_{fA}}{L_{fA}} & 0 & 0 & \frac{-Sw_{A0}}{L_{fA}} \\
0 & \frac{R_{cfB}}{L_{fB}} & 0 & 0 & \frac{1}{L_{fB}} & 0 & 0 & -\frac{R_{cfB}-R_{fB}}{L_{fB}} & 0 & \frac{-Sw_{B0}}{L_{fB}} \\
0 & 0 & \frac{R_{cfC}}{L_{fC}} & 0 & 0 & \frac{1}{L_{fC}} & 0 & 0 & -\frac{R_{cfC}-R_{fC}}{L_{fC}} & \frac{-Sw_{C0}}{L_{fC}} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{-Sw_{A0}}{C_{dc}} & \frac{-Sw_{B0}}{C_{dc}} & \frac{-Sw_{C0}}{C_{dc}} & \frac{-1}{R_{dc}C_{dc}}
\end{bmatrix},$$

$$B = \begin{bmatrix}
\frac{1}{L_{gA}} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{L_{gB}} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{L_{gC}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{-V_{dc0}}{L_{fA}} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fB}} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fC}} \\
0 & 0 & 0 & \frac{i_{A0}}{C_{dc}} & \frac{i_{B0}}{C_{dc}} & \frac{i_{C0}}{C_{dc}}
\end{bmatrix}, \boldsymbol{x}(t) = \begin{bmatrix}
\Delta i_{gA} \\ \Delta i_{gB} \\ \Delta i_{gC} \\ \Delta v_{capA} \\ \Delta v_{capB} \\ \Delta v_{capC} \\ \Delta i_{fA} \\ \Delta i_{fB} \\ \Delta i_{fC} \\ \Delta v_{dc}
\end{bmatrix}, \boldsymbol{u}(t) = \begin{bmatrix}
\Delta v_{pccA} \\ \Delta v_{pccB} \\ \Delta v_{pccC} \\ \Delta sw_A \\ \Delta sw_B \\ \Delta sw_C
\end{bmatrix}$$

$$
\begin{bmatrix} A & k\omega I \\ -k\omega I & A \end{bmatrix} = \left[\begin{array}{cccccccccccccccccccc}
-\frac{-R_{gA}-R_{cfA}}{L_{gA}} & 0 & 0 & \frac{-1}{L_{gA}} & 0 & 0 & \frac{R_{cfA}}{L_{gA}} & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{-R_{gB}-R_{cfB}}{L_{gB}} & 0 & 0 & \frac{-1}{L_{gB}} & 0 & 0 & \frac{R_{cfB}}{L_{gB}} & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{-R_{gC}-R_{cfC}}{L_{gC}} & 0 & 0 & \frac{-1}{L_{gC}} & 0 & 0 & \frac{R_{cfC}}{L_{gC}} & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{C_{fA}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fA}} & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{C_{fB}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fB}} & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{C_{fC}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fC}} & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 & 0 \\
\frac{R_{cfA}}{L_{fA}} & 0 & 0 & \frac{1}{L_{fA}} & 0 & 0 & -\frac{R_{cfA}-R_{fA}}{L_{fA}} & 0 & 0 & \frac{-Sw_{A0}}{L_{fA}} & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 & 0 \\
0 & \frac{R_{cfB}}{L_{fB}} & 0 & 0 & \frac{1}{L_{fB}} & 0 & 0 & -\frac{R_{cfB}-R_{fB}}{L_{fB}} & 0 & \frac{-Sw_{B0}}{L_{fB}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 & 0 \\
0 & 0 & \frac{R_{cfC}}{L_{fC}} & 0 & 0 & \frac{1}{L_{fC}} & 0 & 0 & -\frac{R_{cfC}-R_{fC}}{L_{fC}} & \frac{-Sw_{C0}}{L_{fC}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{-Sw_{A0}}{C_{dc}} & \frac{-Sw_{B0}}{C_{dc}} & \frac{-Sw_{C0}}{C_{dc}} & \frac{-1}{R_{dc}C_{dc}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k\omega \\
-k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{-R_{gA}-R_{cfA}}{L_{gA}} & 0 & 0 & \frac{-1}{L_{gA}} & 0 & 0 & \frac{R_{cfA}}{L_{gA}} & 0 & 0 & 0 \\
0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{-R_{gB}-R_{cfB}}{L_{gB}} & 0 & 0 & \frac{-1}{L_{gB}} & 0 & 0 & \frac{R_{cfB}}{L_{gB}} & 0 & 0 \\
0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{-R_{gC}-R_{cfC}}{L_{gC}} & 0 & 0 & \frac{-1}{L_{gC}} & 0 & 0 & \frac{R_{cfC}}{L_{gC}} & 0 \\
0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{C_{fA}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fA}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{C_{fB}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fB}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{C_{fC}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{C_{fC}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & \frac{R_{cfA}}{L_{fA}} & 0 & 0 & \frac{1}{L_{fA}} & 0 & 0 & -\frac{R_{cfA}-R_{fA}}{L_{fA}} & 0 & 0 & \frac{-Sw_{A0}}{L_{fA}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & \frac{R_{cfB}}{L_{fB}} & 0 & 0 & \frac{1}{L_{fB}} & 0 & 0 & -\frac{R_{cfB}-R_{fB}}{L_{fB}} & 0 & \frac{-Sw_{B0}}{L_{fB}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & \frac{R_{cfC}}{L_{fC}} & 0 & 0 & \frac{1}{L_{fC}} & 0 & 0 & -\frac{R_{cfC}-R_{fC}}{L_{fC}} & \frac{-Sw_{C0}}{L_{fC}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k\omega & 0 & 0 & 0 & 0 & 0 & 0 & \frac{-Sw_{A0}}{C_{dc}} & \frac{-Sw_{B0}}{C_{dc}} & \frac{-Sw_{C0}}{C_{dc}} & \frac{-1}{R_{dc}C_{dc}}
\end{array}\right],
$$

$$
\begin{bmatrix} B & O \\ O & B \end{bmatrix} = \left[\begin{array}{cccccccccccc}
\frac{1}{L_{gA}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{L_{gB}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{L_{gC}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{-V_{dc0}}{L_{fA}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fB}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fC}} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i_{A0}}{C_{dc}} & \frac{i_{B0}}{C_{dc}} & \frac{i_{C0}}{C_{dc}} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{gA}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{gB}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{L_{gC}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fA}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fB}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{-V_{dc0}}{L_{fC}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i_{A0}}{C_{dc}} & \frac{i_{B0}}{C_{dc}} & \frac{i_{C0}}{C_{dc}}
\end{array}\right],
\begin{bmatrix} \boldsymbol{\hat{x}_{R,k}} \\ \boldsymbol{\hat{x}_{I,k}} \end{bmatrix} = \begin{bmatrix}
\Delta\hat{i}_{gA,R,k} \\ \Delta\hat{i}_{gB,R,k} \\ \Delta\hat{i}_{gC,R,k} \\ \Delta\hat{v}_{capA,R,k} \\ \Delta\hat{v}_{capB,R,k} \\ \Delta\hat{v}_{capC,R,k} \\ \Delta\hat{i}_{fA,R,k} \\ \Delta\hat{i}_{fB,R,k} \\ \Delta\hat{i}_{fC,R,k} \\ \Delta\hat{v}_{dc,R,k} \\ \Delta\hat{i}_{gA,I,k} \\ \Delta\hat{i}_{gB,I,k} \\ \Delta\hat{i}_{gC,I,k} \\ \Delta\hat{v}_{capA,I,k} \\ \Delta\hat{v}_{capB,I,k} \\ \Delta\hat{v}_{capC,I,k} \\ \Delta\hat{i}_{fA,I,k} \\ \Delta\hat{i}_{fB,I,k} \\ \Delta\hat{i}_{fC,I,k} \\ \Delta\hat{v}_{dc,I,k}
\end{bmatrix},
\begin{bmatrix} \boldsymbol{\hat{u}_{R,k}} \\ \boldsymbol{\hat{u}_{I,k}} \end{bmatrix} = \begin{bmatrix}
\Delta\hat{v}_{pccA,R,k} \\ \Delta\hat{v}_{pccB,R,k} \\ \Delta\hat{v}_{pccC,R,k} \\ \Delta\hat{sw}_{A,R,k} \\ \Delta\hat{sw}_{B,R,k} \\ \Delta\hat{sw}_{C,R,k} \\ \Delta\hat{v}_{pccA,I,k} \\ \Delta\hat{v}_{pccB,I,k} \\ \Delta\hat{v}_{pccC,I,k} \\ \Delta\hat{sw}_{A,I,k} \\ \Delta\hat{sw}_{B,I,k} \\ \Delta\hat{sw}_{C,I,k}
\end{bmatrix}
$$